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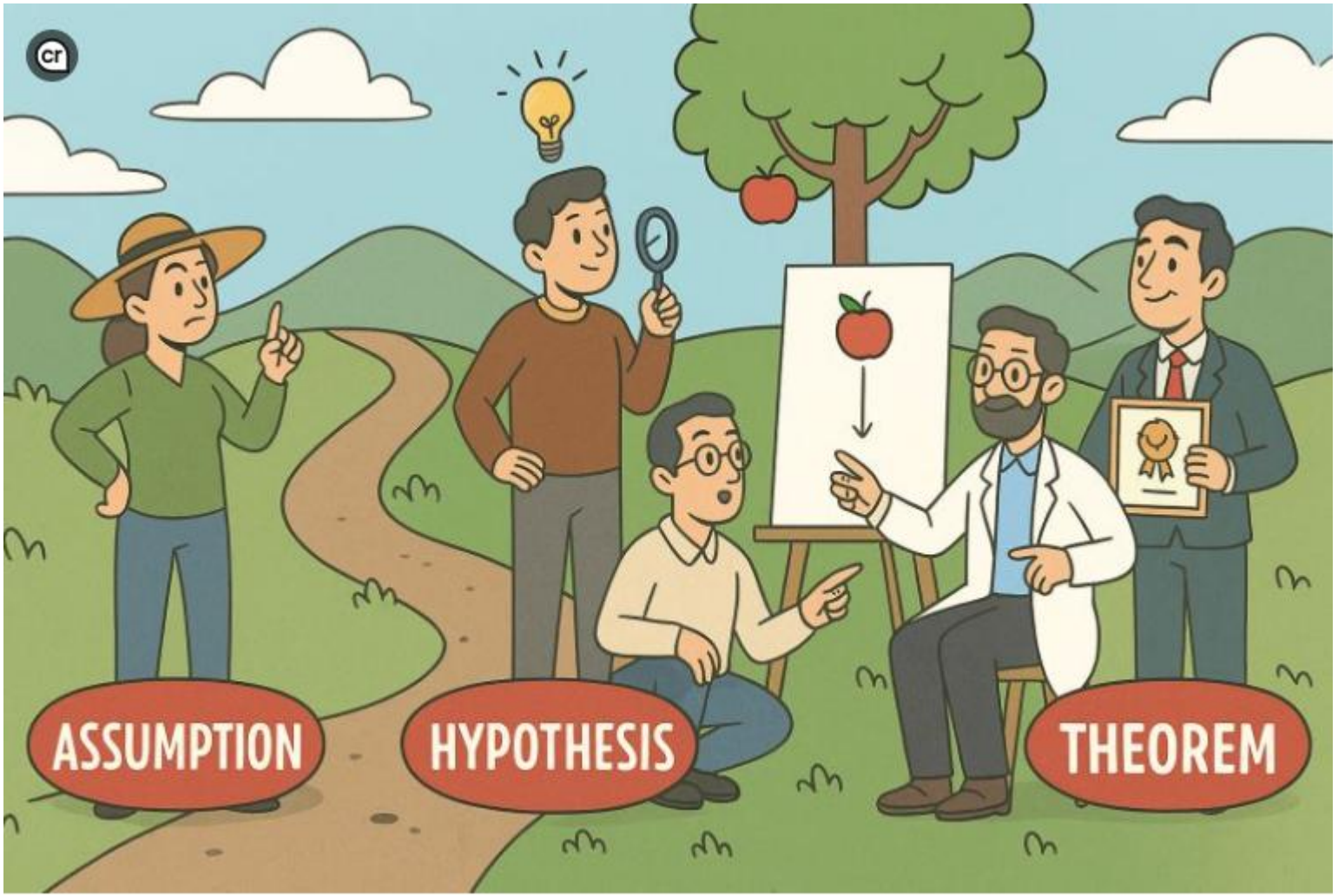
Information and Communication Technology

UNIT - 3
CONCEPT LEARNING

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Contents

- Formulation of Hypothesis
- Probabilistic Approximately Correct Learning
- VC Dimension
- Hypothesis elimination
- Candidate Elimination Algorithm


ASSUMPTION
HYPOTHESIS
THEOREM

Hypothesis

The word hypothesis consists of two words:

Hypo + **thesis** = Hypothesis '**Hypo**' means tentative or subject to the verification and '**Thesis**' means statement about solution of a problem.

The term hypothesis is a **tentative statement about the solution of the problem**. Hypothesis offers a solution of the problem that is to be verified by experiments.

Hypothesis

An accurate definition of a hypothesis is a **tentative explanation** for a **phenomenon**, **observation** or **scientific problem** that can be tested through **further investigation**, **research** and **experimentation**.

Example

"Customers who receive discount coupons make more purchases than customers who do not receive coupons.“

This statement can be tested by analyzing customer purchase data.

Hypothesis

Example 1: Student Performance

Problem: Does studying more improve exam scores?

Hypothesis: Students who study for more hours will score higher marks in examinations.

This prediction can be tested by collecting data on students' study hours and exam scores.

Hypothesis

Example 2: E-Commerce

Problem: Does offering discounts increase sales?

Hypothesis: Offering discounts increases product sales.

A company can test this by comparing sales before and after offering discounts.

Hypothesis

Example 3: Healthcare

Problem: Does regular exercise improve health?

Hypothesis: People who exercise regularly have lower blood pressure than those who do not.

This can be verified using health data.

Applications of Hypothesis in Data Science

Hypotheses are commonly used in:

a) Business analytics

b) Healthcare research

c) Marketing campaigns

d) Banking and finance

e) Education

f) Machine learning model evaluation

Assumption

An **assumption** is a belief or condition that is **accepted as true without testing** or **sufficient evidence**. It provides a starting point for analysis or model development.

Example

A company assumes that:

"Customers prefer online shopping over offline shopping."

This statement is accepted initially but has not yet been verified using data.

Difference Between Hypothesis and Assumption in Data Science

In Data Science, both **hypothesis** and **assumption** are used during **problem-solving** and **data analysis**, but they **serve different purposes**.

A **hypothesis** is a testable statement that is **verified using data and statistical methods**.

An **assumption** is something accepted as **true without immediate proof**.

Basis	Hypothesis	Assumption
Definition	A **testable statement** or prediction about variables.	A **belief** or **condition** accepted as true **without proof**.
Purpose	To be **tested** and **verified** using data.	To provide a **starting point** for analysis or modeling.
Evidence Required	**Requires data** and statistical analysis.	**Does not** require immediate evidence.
Testing	**Always tested** before drawing conclusions.	Usually **not tested initially**.

Basis	Hypothesis	Assumption
Nature	Scientific and evidence-based.	Logical belief or expectation.
Outcome	Can be accepted or rejected after analysis.	May later be validated or modified.
Role in Data Science	Used in hypothesis testing and research.	Used while designing models or defining project conditions.
Reliability	Depends on experimental results.	Based on prior knowledge or experience.

Characteristics of a Good Hypothesis

A good hypothesis should be:

1. **Clear:** Easy to understand.
2. **Specific:** Focused on a particular problem.
3. **Testable:** Can be verified using data or experiments.
4. **Measurable:** Involves measurable variables.
5. **Logical:** Based on facts, observations or previous research.

Importance of Hypothesis

A hypothesis helps to:

- Define the research problem.
- Guide data collection and analysis.
- Support evidence-based decision-making.
- Test assumptions scientifically.
- Draw meaningful conclusions from data.

Formulation of Hypothesis

The Formulation of Hypothesis is an important step in Data Science and statistical analysis. Before analyzing data, a data scientist develops a hypothesis—**a testable statement** or **assumption about the relationship between variables**. This hypothesis is then tested using data to determine whether it is supported by evidence.

Example: a company may assume that **offering discounts increases product sales**. This assumption becomes a hypothesis that can be tested using historical sales data.

Why is Hypothesis Formulation Important?

Hypothesis formulation helps data scientists to:

- ✓ Define the research problem clearly.
- ✓ Guide data collection and analysis.
- ✓ Make objective decisions based on evidence.
- ✓ Validate assumptions using statistical methods.
- ✓ Improve the reliability of conclusions.

Steps in Formulation of Hypothesis

Identify the Problem

↓

Collect the Background Information

↓

Develop a Hypothesis

↓

Collect Relevant Data

↓

Test the Hypothesis

↓

Accept or Reject the Hypothesis

1. Identify the Problem

The first step is to **clearly define the problem or question** that needs to be solved. Without a clear problem, the research has no direction.

Questions to Ask

1. What problem needs to be solved?
2. What do we want to know?
3. Why is this problem important?

1. Identify the Problem

Example: The principal notices that many students are getting low marks in examinations.

Problem: Does studying for more hours improve students' exam performance?

This is the research problem.

2. Collect Background Information

Collect all available information related to the problem before making any prediction. This helps us understand the problem better.

Sources of Information

- ✓ Previous research papers
- ✓ Books
- ✓ School records
- ✓ Internet
- ✓ Surveys
- ✓ Expert opinions

2. Collect Background Information

Example

The principal collects -

1. Previous exam results
2. Students' study hours
3. Teacher observations

He finds that students who studied longer generally scored higher marks.

2. Collect Background Information

Purpose

Background information helps us understand:

- ✓ Existing knowledge
- ✓ Possible reasons
- ✓ Important variables

3. Develop a Hypothesis

A hypothesis is a prediction that can be tested using data. It predicts the relationship between variables.

Types of Hypothesis

A**. Null Hypothesis (H_0):** There is **no relationship** between variables.

Example: Studying more hours **does not improve** exam marks.

B**. Alternative Hypothesis (H_1):** There **is a relationship** between variables.

Example: Studying more hours **improves** exam marks.

Purpose: The hypothesis gives direction to the research.

4. Collect Relevant Data

Now collect data that will help test the hypothesis.

The data should be:

1. Accurate
2. Reliable
3. Relevant

4. Collect Relevant Data

Example: The principal collects the following data.

Student	Study Hours	Exam Marks
Rahul	2	55
Priya	3	60
Aman	5	75
Neha	6	82
Ravi	7	90

Observation: Students who studied more hours generally obtained higher marks.

5. Test the Hypothesis

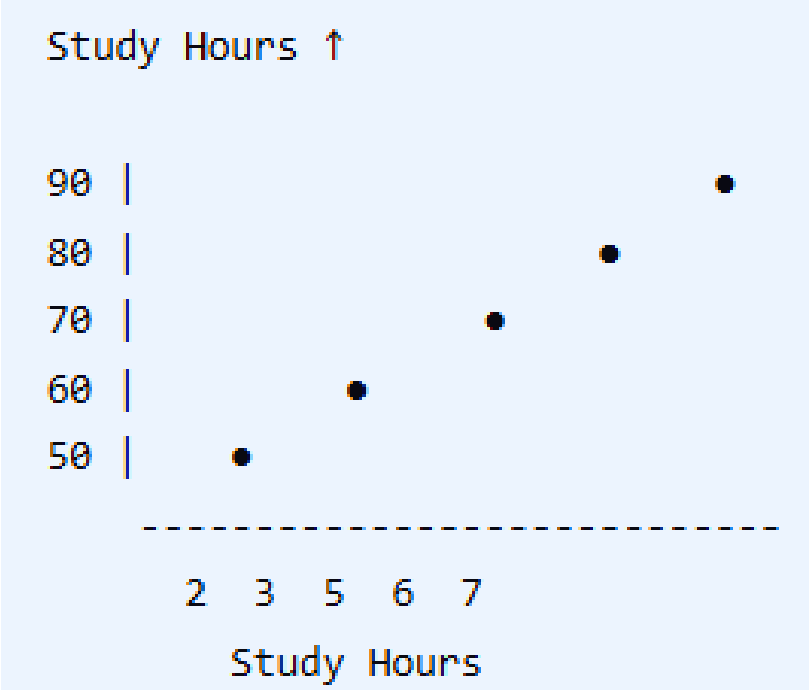
Analyze the collected data using statistical or machine learning techniques.

The goal is to determine whether the data supports the hypothesis.

Example

The principal creates a graph.

The graph shows that exam marks increase as study hours increase. Statistical analysis also confirms a positive relationship.



5. Test the Hypothesis

Common Statistical Tests

a) Correlation Analysis

b) Regression Analysis

c) t-Test

d) ANOVA

e) Chi-Square Test

6. Accept or Reject the Hypothesis

Based on the analysis, decide whether the hypothesis is supported.

Case 1: Accept the Alternative Hypothesis - If data shows a strong relationship.

Result: Students studying longer score better marks.

Decision: Reject H_0 and Accept H_1

6. Accept or Reject the Hypothesis

Based on the analysis, decide whether the hypothesis is supported.

Case 2: Accept the Null Hypothesis - If no significant relationship exists.

Example: Suppose analysis shows:

Students studying 2 hours and 8 hours get almost the same marks.

Then, Do not reject H_0.

Types of Hypothesis

There are several types of hypotheses depending on the machine learning problem-

1. Simple Hypothesis
2. Complex Hypothesis
3. Linear Hypothesis
4. Non-Linear Hypothesis
5. Specific Hypothesis
6. General Hypothesis
7. Null Hypothesis (H_0)

Types of Hypothesis

There are several types of hypotheses depending on the machine learning problem-

8. Alternative Hypothesis (H_1)
9. Conjunctive Hypothesis
10. Disjunctive Hypothesis

Types of Hypothesis

1. **Simple Hypothesis:**

A simple hypothesis considers **only one input feature** to predict the output.

Example: Student Result Prediction
Only Attendance is considered.

Attendance	Result
90%	Pass
60%	Fail

Hypothesis

If **Attendance > 75%**
Pass
Else
Fail

Only one attribute is used.

Types of Hypothesis

2. **Complex Hypothesis:**

A complex hypothesis **considers multiple input features (variables)**.

Example: Student Result

Hypothesis

Attendance	Assignment	Internal Marks	Result
90	Yes	40	Pass
70	Yes	35	Pass
60	No	20	Fail

Types of Hypothesis

2. **Complex Hypothesis:**

Hypothesis

IF **Attendance >75** AND **Assignment = Yes** AND **Internal Marks >30**
    Pass
ELSE
    Fail

Types of Hypothesis

3. **Linear Hypothesis:**

The output is represented using a **linear equation**.

Example: House Price Prediction

Used in: Linear Regression

Types of Hypothesis

4. **Non-Linear Hypothesis:**

The relationship between input and output is **not linear**.

Example: Disease Prediction

Age increases risk slowly at first and rapidly later.

Hypothesis: Risk = Age^2

Instead of : Risk = Age

Algorithms: Decision Tree, Random Forest, Neural Network

Types of Hypothesis

5. **Specific Hypothesis:**

A specific hypothesis matches **very few training examples**.

Example: Fruit Dataset

Specific Hypothesis

Color = Red

Shape = Round

Fruit = Apple

Only this exact combination works.

Color	Shape	Fruit
Red	Round	Apple

Types of Hypothesis

6. **General Hypothesis:**

A general hypothesis covers **many possible examples**.

Example: Fruit Dataset

Any Round Fruit → Apple

OR

Color = Any

Shape = Round

This covers many fruits.

Color	Shape	Fruit
Red	Round	Apple

Types of Hypothesis

7. **Null Hypothesis (H_0) :**

There is **no relationship** between variables.

Example: Student attendance has **no effect** on exam marks.

Symbol - H_0

Used in

- Hypothesis Testing
- Statistics

Types of Hypothesis

8. **Alternative Hypothesis (H_1):**

There is **a relationship** between variables.

Example: Attendance improves student performance.

Symbol – H_1

Types of Hypothesis

9. **Conjunctive Hypothesis:**

Uses **AND** operator.

Example:

Attendance >75 AND Assignment = Yes

THEN **Pass**

Both conditions must be true.

Types of Hypothesis

10. **Disjunctive Hypothesis:**

Uses **OR** operator.

Example:

Attendance >75 OR Internal Marks >40

THEN **Pass**

Only one condition needs to be true.

Example of Hypothesis

Real-Life Example (Loan Approval)

Suppose a bank wants to approve loans.

Features: Income, Credit Score, Employment Status

Simple Hypothesis: If Income > ₹50,000 → Approve Loan.

Complex Hypothesis: If Income > ₹50,000 **AND** Credit Score > 750 **AND** Employment = Permanent → Approve Loan.

Null Hypothesis (H_0): Income has no effect on loan approval.

Alternative Hypothesis (H_1): Higher income increases the chance of loan approval.

Probabilistic Approximately Correct (PAC) Learning

Probabilistic Approximately Correct **(PAC)** Learning is a theoretical framework in Machine Learning introduced by **Leslie Valiant** in **1984**.

It explains **how much data is needed for a machine learning model to learn a concept with high confidence and acceptable accuracy**.

OR

A machine learning model does not need to be 100% correct.

The "**probably**" aspect refers to the **confidence level of the algorithm**, while the "**approximately correct**" aspect refers to the **accuracy of the hypothesis**.

Probabilistic Approximately Correct (PAC) Learning

PAC theorem

Mathematically, the PAC learning theorem can be expressed as:

$$m \geq \frac{1}{\varepsilon}\left(\ln|H| + \ln\frac{1}{\delta}\right)$$

where

- ✓ m = number of training samples
- ✓ |H| = hypothesis space size
- ✓ ε = desired accuracy
- ✓ δ = confidence level

Probabilistic Approximately Correct (PAC) Learning

Example

Suppose

hypothesis space size $|H| = 100$

desired accuracy $\varepsilon = 0.05$

confidence level $\delta = 0.01$ then

ln(100)=4.605

ln(100)=4.605

Therefore,

Probabilistic Approximately Correct (PAC) Learning

Example

$$\ln(|H|) = \ln(100) = 4.605$$

$$\ln\left(\frac{1}{\delta}\right) = \ln\left(\frac{1}{0.01}\right) = \ln(100) = 4.605$$

$$m \geq \frac{1}{0.05}(4.605 + 4.605) = 20 \times 9.21 = 184.$$

$$100 = 10^2$$

$$\ln(10^2) = 2\ln(10)$$

$$\ln(10) \approx 2.302585$$

$$\ln(100) = 2 \times 2.302585$$

$$= 4.605170$$

$$\boxed{\ln(100) \approx 4.605}$$

Probabilistic Approximately Correct (PAC) Learning

Example

Approximately 185 examples are sufficient under the PAC bound.

VC Dimension (Vapnik–Chervonenkis Dimension)

The VC is a measure of the **capacity (complexity)** of a machine learning model. It indicates **how powerful a model is in learning different patterns from data**.

VC Dimension is the maximum number of data points that a model can classify correctly in every possible way.

VC Dimension (Vapnik–Chervonenkis Dimension)

It helps answer questions such as:

- How complex is the model?
- Will the model work well to new data?
- Is the model likely to overfit?

VC Dimension is widely used in **Statistical Learning Theory** and **PAC (Probably Approximately Correct) Learning**.

Why did an ML Model need VC Dimension?

This is a very **important question**. Let's understand it through a story.

The Real Problem — "Model gave 100% in Training, then why did it fail in the Real World?"

Imagine you taught a Student.

Training = Give them old Question Papers

Exam = New Questions came → Student failed

This is exactly what happens in ML — this is called **Over fitting**.

What was the Problem before?

In the old days, Scientists had **no Mathematical way to know** that:

"This Model is good on Training Data — but how good will it be on new Data?"

Everything was just guesswork. There was no Guarantee.

Then Vapnik and Chervonenkis thought 💡

In 1971 they asked —

"If we can mathematically measure the Complexity of a Model, then we can also predict how it will perform on new Data!“

And from this, **VC Dimension** was born.

Then Vapnik and Chervonenkis thought 💡

VC Dimension solved 3 Big Problems

Problem 1 — How **"powerful"** is the Model?

Earlier there was no way to know which of two models was more capable.

VC Dimension gave a **single number.**

Then Vapnik and Chervonenkis thought 💡

VC Dimension solved 3 Big Problems

Problem 2 — **How much Data** is needed?

From VC Dimension this Formula came out:

Larger the VC Dimension

↓

More Training Data is needed

↓

Otherwise the Model will Overfit

Then Vapnik and Chervonenkis thought 💡

VC Dimension solved 3 Big Problems

Problem 3 — When to **Trust** a Model?

VC Dimension gave a mathematical guarantee:

"If your VC Dimension is low and Data is more — then the model's real-world error will remain bounded."

That means now there was **no guesswork, but a Proof.**

Understand with one more Real Life Example

Situation	Meaning
Small Child (Low VC Dim)	Can only learn Simple rules — Underfit
PhD Scholar (High VC Dim)	Can learn Complex patterns — but if given fewer examples then will Overfit
Good ML Model	Right Balance of VC Dim and Data

VC Dimension in Data Science

VC Dimension is useful in:

- ✓ Model selection
- ✓ Understanding over fitting and under fitting
- ✓ Statistical learning theory
- ✓ PAC learning
- ✓ Estimating the amount of training data required
- ✓ Comparing hypothesis classes
- ✓ Evaluating model capacity

Advantages of VC Dimension

1. Measures model complexity mathematically.
2. Explains why complex models need more data.
3. Helps estimate sample complexity.
4. Provides a theoretical basis for generalization.

Limitations of VC Dimension

1. Difficult to calculate exactly for many modern models (especially deep neural networks).
2. A higher VC Dimension does not always mean better real-world performance.
3. It is a theoretical concept that does not directly consider optimization techniques or the quality of data.

Hypothesis Elimination

The Candidate Elimination Algorithm is a supervised machine learning algorithm introduced by **Tom M. Mitchell**. Hypothesis Elimination is a concept in Machine Learning used to remove inconsistent hypotheses from the hypothesis space based on training examples.

It helps in identifying the set of hypotheses that correctly classify all observed training examples.

Hypothesis Elimination

Key concept

- Start with all possible hypotheses
- Eliminate hypotheses inconsistent with data
- Retain only valid hypotheses

Hypothesis Elimination

Applications

- Concept Learning
- Classification Problems
- Machine Learning Training Models

Hypothesis Elimination

Concept Learning:

It is the process by which a machine learns a **general rule (concept)** from a set of labeled training examples. The goal is to determine whether new, unseen examples belong to a particular target concept.

In simple terms, concept learning teaches a computer **how to classify objects into "Yes" or "No" categories** based on examples.

Hypothesis Elimination

Concept Learning:

It is the process by which a machine learns a **general rule (concept)** from a set of labeled training examples. The goal is to determine whether new, unseen examples belong to a particular target concept.

In simple terms, concept learning teaches a computer **how to classify objects into "Yes" or "No" categories** based on examples.

Hypothesis Elimination – Candidate Elimination Algorithm

Concept Learning:

Real-Life Example- Imagine you want to teach a child to recognize apples.

You show the child several fruits:

Fruit	Color	Shape	Sweet	Apple?
Apple	Red	Round	Yes	✓ Yes
Apple	Green	Round	Yes	✓ Yes
Banana	Yellow	Long	Yes	✗ No
Orange	Orange	Round	Yes	✗ No

Hypothesis Elimination

Concept Learning:

Real-Life Example-

From these examples, the child learns the concept:

"An apple is generally round and can be red or green."

Similarly, a machine learns concepts from data.

Hypothesis Elimination – Candidate Elimination Algorithm

Hypothesis Space and Version Space

Hypothesis Space (H) : The **complete set of all possible hypotheses** that can describe the training data.

Version Space (VS): The **subset of hypotheses** that are consistent with all training examples.

Hypothesis Elimination – Candidate Elimination Algorithm

Hypothesis Space and Version Space

Important Components:

1. Specific Boundary (S)

Represents the **most specific hypothesis** that is consistent with the training data.

$S=\{\langle \emptyset,\emptyset,\emptyset,\emptyset \rangle\}$

Example: S = (Ø, Ø, Ø, Ø)

Nothing is accepted initially.

Hypothesis Elimination – Candidate Elimination Algorithm

Hypothesis Space and Version Space

Important Components:

2. General Boundary (G)

Represents the **most general hypothesis**.

G = (?, ?, ?, ?)

Everything is accepted initially.

Hypothesis Elimination – Candidate Elimination Algorithm

Hypothesis Space and Version Space

Important Components:

3. Version Space

The hypotheses lying between **S** and **G** are called the **Version Space**.

Most Specific -------- Version Space -------- Most General

S **G**

Version Space contains **all hypotheses consistent with the training data**.

Hypothesis Elimination – Candidate Elimination Algorithm

Symbols Used

Symbol	Meaning
Ø	No value assigned
?	Any value is acceptable
Yes	Positive example
No	Negative example

Working of Candidate Elimination Algorithm

What is a Hypothesis?

A **hypothesis (H)** is a rule that predicts the output **(target class)** based on the input attributes.

Example

Suppose we want to predict whether a person will play tennis.

Working of Candidate Elimination Algorithm

Example

Suppose we want to predict whether a person will play tennis.

Weather	Temp	Humidity	Wind	Play
Sunny	Warm	Normal	Strong	Yes
Sunny	Warm	High	Strong	Yes
Rainy	Cold	High	Strong	No
Sunny	Warm	High	Weak	Yes

Working of Candidate Elimination Algorithm

Example

Step 1 -

Initialize

S = (Ø, Ø, Ø, Ø)

G = (?, ?, ?, ?)

Working of Candidate Elimination Algorithm

Step 2 -

For every training example

If the example is **Positive**

Generalize **S**

Remove inconsistent hypotheses from **G**

If the example is **Negative**

Specialize **G**

Remove inconsistent hypotheses from **S**

Working of Candidate Elimination Algorithm

Training Example 1

Sunny Warm Normal Strong → Yes [positive]

Generalize S

S = (Sunny, Warm, Normal, Strong)

G remains

G = (?, ?, ?, ?)

Working of Candidate Elimination Algorithm

Training Example 2

Sunny Warm High Strong → Yes (Humidity changed.)

Generalize S

S = (Sunny, Warm, ?, Strong)

G remains

G = (?, ?, ?, ?)

Working of Candidate Elimination Algorithm

Training Example 3

Rainy Cold High Strong → No [Negative example]

Specialize G - Possible hypotheses

(Sunny, ?, ?, ?)

(?, Warm, ?, ?)

(?, ?, Normal, ?)

Remove inconsistent ones.

Working of Candidate Elimination Algorithm

Training Example 4

Sunny Warm High Weak → Yes (Wind changes.) [Positive]

Generalize S

S = (Sunny, Warm, ?, ?)

Final G

(Sunny, ?, ?, ?)

(?, Warm, ?, ?)

Working of Candidate Elimination Algorithm

Final Version Space

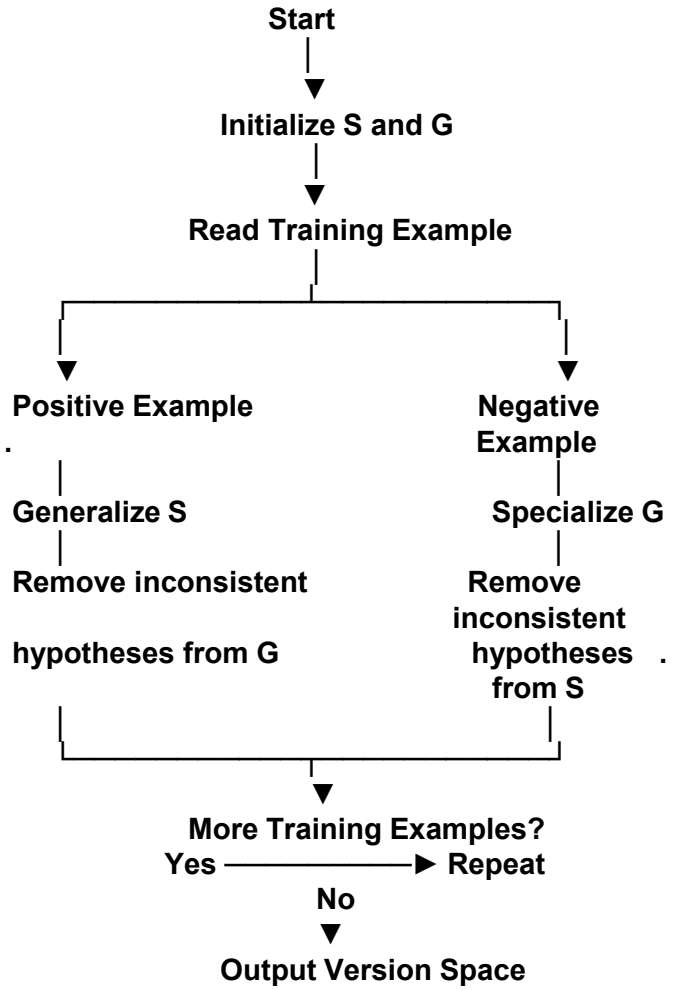
S = (Sunny, Warm, ?, ?)

G = (Sunny, ?, ?, ?)

(?, Warm, ?, ?)

These hypotheses are consistent with all training example

Candidate Elimination Flowchart



Advantages

I. Finds **all consistent hypotheses**, not just one.

II. Eliminates inconsistent hypotheses systematically.

III. Helps understand the **version space** of possible solutions.

IV. Useful for studying concept learning and hypothesis spaces.

Disadvantages

I. Computationally expensive for large hypothesis spaces.

II. Sensitive to noisy or mislabeled data.

III. Not suitable for very large real-world datasets. Assumes training data is noise-free.

Applications

- ✓ Concept Learning
- ✓ Machine Learning
- ✓ Education Rule-Based Systems
- ✓ Knowledge Discovery
- ✓ Medical Diagnosis
- ✓ Pattern Recognition
- ✓ Classification Problems

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